

# EMMA'S ATLAS OF NUMBERS AND SHAPES

- A NOT-SO-BRIEF GUIDE TO ALL THE MATH I KNOW -

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**Part I**

**Foundations**

## **Chapter 1**

# **An Introduction to Mathematical Reasoning**

# Chapter 2

## Sets

Almost all of modern mathematics can be put in the context of the study of *sets* and their *elements*. Generally, given a collection of objects  $a, b, c, d, \dots$ , they can be taken together as a set, generally written in curly brackets and represented by a capital letter:

$$M := \{a, b, c, d, e, \dots\}$$

However, an approach where any collection of mathematical objects forms a set quickly runs into trouble - famously, *Russel's paradox* asks:

*"Let  $R$  be the set of all sets that don't contain themselves. Does  $R$  contain itself?"*

If  $R$  doesn't contain itself, then by definition it must contain itself, but if it contains itself, then it cannot contain itself! If  $R \in R$  is true, then it must be false! We arrive at a classic paradox. The conclusion, then, is that the set of all sets that don't contain themselves cannot exist - we cannot define sets to be any collection of things defined by any property, and must instead establish ground rules about which sets exist and exactly how we can use sets we already know to construct new ones.

These ground rules are given in the form of *axioms*, which are statements that mathematicians take for granted, which can then be used as a starting point to then derive "the rest of mathematics" from. The following chapters contain a brief and hopefully easily understandable introduction to the ZFC axiom system, which was developed in the early 20th century by Ernst Zermelo and Abraham Fraenkel and is now the most widely accepted axiomatization of mathematics.

### 2.1 Infinity: $\emptyset$ and $\mathbb{N}$

### 2.2 Extensionality

To say " $b$  is an element of  $M$ ", or equivalently " $b$  contains  $M$ ", we simply write:

$$b \in M$$

One of the most basic axioms of set theory is the *Axiom of Extensionality*, which states:

### Axiom 2.1: Axiom of Extensionality

Two sets  $A$  and  $B$  are equal if and only if they contain the same elements.

The Axiom of Extensionality establishes that sets are uniquely defined by the objects they contain and have no additional structure. In particular:

1. The order of elements in a set doesn't matter.  $\{1, 2\}$  and  $\{2, 1\}$  are the same set.
2. The number of times an element appears within the same set doesn't matter.  $\{1\}$ ,  $\{1, 1\}$ , and  $\{1, 1, 1\}$  are the same set.

### Definition 2.2: Subsets

We call a set  $A$  a **subset** of a set  $B$  if and only if every element of  $A$  is contained in  $B$ . We write " $A$  is a subset of  $B$ " as  $A \subseteq B$ .

The demand that "every element of  $A$  is contained in  $B$ " can be stated more formally as " $x \in A$  implies  $x \in B$ ". Importantly, this means that every set is a subset of itself - we always have  $A \subseteq A$ .

This definition gives us another way of phrasing the Axiom of Extensionality - two sets are equal if and only if each is a subset of the other:

$$A = B \Leftrightarrow A \subseteq B \text{ and } B \subseteq A$$

This idea is incredibly important for writing out proofs in practice - if a problem asks you to establish that two sets  $A$  and  $B$  are the same, then the easiest approach is almost always to show that any element of  $A$  must be contained in  $B$ , and to then show that any arbitrary element of  $B$  must be contained in  $A$ .

The subset relation is frequently written as  $A \subset B$ . However, other authors reserve  $A \subset B$  to mean " $A$  is a subset of  $B$  that isn't equal to  $B$ " - we call such a set a **proper subset** of  $B$ . I will therefore attempt to only use the symbols  $\subseteq$  and  $\subsetneq$ , which clear up this ambiguity.

- 2.3 Specification**
- 2.4 Pairing, Union, and Power Sets**
- 2.5 Relations**
- 2.6 Functions**
- 2.7 Replacement**
- 2.8 Regularity and Well-Foundedness**
- 2.9 Choice**
- 2.10 Cardinality**

## **Part II**

# **Foundations of Analysis**

# Chapter 1

## The Real Numbers $\mathbb{R}$

### 1.1 Ordered Algebraic Structures

### 1.2 Dedekind Cuts

### 1.3 Uniqueness of $\mathbb{R}$

### 1.4 Cardinality of $\mathbb{R}$

## Chapter 2

# The Concept of Distance

### 2.1 Metric Spaces

A metric space is a set, alongside with a *distance function*, more commonly known as a *metric*,  $d$  which takes two points and outputs a distance between the points. Naturally, we want this "distance" to actually give us some usable structure on the set.

Now, what pieces of geometric intuition should we bake into our notion of distance? I would argue that the most fundamental use of "distance" is to be able to *compare distances between any two pairs of points*. Therefore, our metric should have as its output a *totally ordered set*.

We also want to be able to add distances together as a way of comparing them - logically, the distance between two points  $x$  and  $y$  plus the distance between  $y$  and  $z$  gives us an upper bound to the distance between  $x$  and  $z$ , since we can always take this path to get from  $x$  to  $z$ , but there might be a shorter path we haven't found yet. As a formula, this is the *triangle inequality*

$$d(x, z) \leq d(x, y) + d(y, z)$$

Finally, if we have an ordered group structure, we would really like for our order to be archimedean - the distance between two distinct points should be greater than zero, and it shouldn't be infinitesimal.

This leaves  $\mathbb{R}_{\geq 0}$  as the obvious candidate for the domain of our distance function, since any archimedean ordered group is a subgroup of  $\mathbb{R}_{\geq 0}$  anyways. This shouldn't be too much of a surprise, even without this logical background - I assume that when most people read "distance between two points", they intuitively already assume that these distances will be given by positive real numbers.

### Definition 2.1: Metric Spaces

A *metric space* is a set  $M$  together with a function  $d : M \times M \rightarrow \mathbb{R}_{\geq 0}$ , such that:

1.  $d$  is *positive definite*, i.e.  $d(x, y) = 0$  if and only if  $x = y$ .
2.  $d$  is *symmetric*, i.e.  $d(x, y) = d(y, x)$
3.  $d$  fulfills the *triangle inequality*

$$d(x, z) \leq d(x, y) + d(y, z)$$

**Theorem 2.2.** Let  $M^n := \prod_{i=1}^n M$  be a finite cartesian product of metric spaces. Then the *Euclidean distance function*

$$d_\infty(x, y) := \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

gives a metric on  $M^n$ .

**Theorem 2.3.** Let  $M^n := \prod_{i=1}^n M$  be a finite cartesian product of metric spaces. Then the *Chebyshev distance function*

$$d_\infty(x, y) := \max_{i \in [1..n]} d(x_i - y_i)$$

gives a metric on  $M^n$ .

### Definition 2.4: Balls

Let  $(M, d)$  be a metric space. Then the *open ball of radius  $r$  around a point  $x$*  is the set

$$B_r(x) := \{y \in M : d(x, y) < r\},$$

and the *closed ball of radius  $r$  around  $x$*  is the set

$$\overline{B}_r(x) := \{y \in M : d(x, y) \leq r\},$$

### Definition 2.5: Open and Closed Sets

Let  $(M, d)$  be a metric space. Then we call a subset  $U \subseteq M$  *open* (in  $M$ ) if it contains an open ball (of any radius) around all of its points, i.e.

$$\forall x \in U :$$

$$\exists r \in \mathbb{R}_{\geq 0} :$$

$$B_r(x) \subseteq U.$$

We call  $U$  *closed* if its complement  $M \setminus U$  is open.

Intuitively, a set is open if it contains "a little bit of wiggle room" around any point contained in it. This means that sets with a "clear border", such as our closed balls, aren't open. For example, the interval  $[0, 1] \subset \mathbb{R}$  doesn't leave any wiggle room around 0 and 1.

Importantly, while it may seem unintuitive, there are in fact sets that are both closed and open at the same time. We call such sets *clopen*.

**Exercise 2.6.** *There are exactly two clopen subsets of  $\mathbb{R}$ . What are they?*

**Definition 2.7.** Let  $(M, d)$  be a metric space. We call a subset  $N \subseteq M$  *dense in  $M$*  if every nonempty open subset of  $M$  contains an element of  $N$ .

**Lemma 2.8.** Every nonempty open subset of  $\mathbb{R}$  is uncountable.

*Proof.* 1. Since every nonempty open subset of  $\mathbb{R}$  contains an open interval  $(a, b)$ , it suffices to show that open intervals are uncountable.

2. The map

$$\begin{aligned}\varphi_1 : (a, b) &\rightarrow \left(-\frac{b-a}{2}, \frac{b-a}{2}\right) \\ x &\mapsto x - \frac{b+a}{2}\end{aligned}$$

is a bijection.

3. The map

$$\begin{aligned}\varphi_2 : \left(-\frac{b-a}{2}, \frac{b-a}{2}\right) &\rightarrow (-1, 1) \\ x &\mapsto \frac{2x}{b-a}\end{aligned}$$

is also a bijection.

4. Finally, the map

$$\begin{aligned}\varphi_3 : (-1, 1) &\rightarrow \mathbb{R} \\ x &\mapsto \frac{x}{1-|x|}\end{aligned}$$

is a bijection (the inverse is the map we get by replacing the  $-$  with a  $+$ ).

5. Therefore,  $\varphi_3 \circ \varphi_2 \circ \varphi_1$  is a bijection  $(a, b) \cong \mathbb{R}$ , so any open interval is uncountable, so any open subset of  $\mathbb{R}$  is uncountable.  $\square$

### Theorem 2.9

The set of rational numbers  $\mathbb{Q}$  is dense in  $\mathbb{R}$ . The set of irrational numbers  $\mathbb{R} \setminus \mathbb{Q}$  is also dense in  $\mathbb{R}$ .

*Proof.* 1. Since any nonempty open subset  $U \subseteq \mathbb{R}$  contains an open interval  $(a, b)$ , it suffices to show that any open interval contains a rational number. We assume without loss of generality that  $0 \leq$

$a < b$ , since  $a < 0 < b$  already implies  $0 \in (a, b)$ , while  $a < b \leq 0$  implies that  $(a, b)$  contains a rational number if and only if  $(-b, -a)$  contains a rational number.

Now, since the ordering on the reals is archimedean, we can find a natural number  $n \in \mathbb{N}$  such that  $\frac{1}{n} < b - a$ . Now, the set

$$L = \left\{ k \in \mathbb{N} \mid \frac{k}{n} > a \right\}$$

is nonempty - if  $a = 0$ , then  $1 \in M$ , and if  $a \neq 0$ , then the archimedean property tells us that there exists a  $k \in \mathbb{N}$  such that

$$\frac{1}{k} < \frac{1}{na},$$

which implies  $\frac{k}{n} > a$ . Since  $M$  is a nonempty subset of  $\mathbb{N}$ , there must be a smallest element  $m$ . Since  $m$  is minimal, we have  $m - 1 \notin M$ , which implies

$$\frac{m-1}{n} \leq a,$$

which in turn implies

$$\frac{m}{n} = \frac{m-1}{n} + \frac{1}{n} < a + (b-a) = b$$

Therefore,  $\frac{m}{n}$  is a rational number contained in  $(a, b)$ , so  $\mathbb{Q}$  is dense in  $\mathbb{R}$ .

2. Since any open subset  $U \subseteq \mathbb{R}$  is uncountable, while the rationals are countable,  $U$  must contain an uncountable number of irrationals. Therefore,  $\mathbb{R} \setminus \mathbb{Q}$  is dense in  $\mathbb{R}$ .

□

### Corollary 2.10

The set of rational numbers  $\mathbb{Q}$  is neither closed nor open in  $\mathbb{R}$ .

*Proof.* Since both sets are dense in  $\mathbb{R}$ , every open ball containing a rational number must also contain an irrational number, and vice versa. □

## 2.2 Topologies

Topological spaces generalize metric spaces. By inheriting the concept of open and closed sets from metric spaces, they are able to still capture notions of closeness without putting a numerical value on exactly how close any given pair of points are. The notion of a topology is based on the following insight:

**Exercise 2.11.** Let  $(M, d)$  be a metric space. Then:

1. The empty set is open in  $M$ .
2. Any union of sets that are open in  $M$  is open in  $M$ .
3. Any finite intersection of sets that are open in  $M$  is open in  $M$ .
4.  $M$  is open in  $M$ .

We take this idea and make it a definition:

**Definition 2.12**

Let  $X$  be a set. Then a **topology on  $X$**  is a set  $\mathcal{O}$  of subsets of  $X$ , called **open sets**, such that:

1. The empty set is open.
2. Any union of open sets is open.
3. Any finite intersection of open sets is open.
4.  $X$  is open.

Most of the topological spaces we care about in practice are already metric spaces, but many proofs flow more elegantly when phrased in the generalized language of topological spaces, and there *are* topological spaces that aren't metric spaces.

**Theorem 2.13: Extremal Topologies**

Let  $X$  be an arbitrary set. Then the following two sets of subsets are always topologies on  $X$ :

1. The **indiscrete topology**, where the only open sets are  $X$  and the empty set.
2. The **discrete topology**, where every subset is open.

*Proof Sketch.* Both topologies fulfill all requirements trivially. :) "□"

**Definition 2.14.** We say that a metric  $d$  on a space  $M$  **generates** a given topology  $\mathcal{O}$  on  $M$  if they give the same open sets, i.e. the open sets in  $\mathcal{O}$  are exactly the open sets in  $(M, d)$ .

**Exercise 2.15.**

1. Show that the discrete metric generates the discrete topology.
2. Show that there is no metric generating the indiscrete topology.

Two different metrics can generate the same topology:

**Exercise 2.16.** Show that the euclidean metric also generates the discrete topology on  $\mathbb{N}$ .

**Exercise 2.17.** Let  $d$  be a metric on a set  $M$ . Let  $c \in \mathbb{R}_{>0}$ .

1. Show that

$$d_c(x, y) = c \cdot d(x, y)$$

is a metric on  $M$ .

2. Show that  $d$  and  $d_c$  generate the same topologies on  $M$ .

This fact can be seen as a formalization of the intuition that changing the specific unit we choose to measure space with doesn't change its underlying structure.

#### Definition 2.18: Interior, Exterior and Boundary

Let  $X$  be a topological space. Let  $A \subseteq X$ . Then:

1. The **interior**  $\text{int } A$  of  $A$  is the set of all points that have an open neighbourhood entirely contained in  $A$ .
2. The **exterior**  $\text{ext } A$  of  $A$  is the interior of the complement  $X \setminus A$ , i.e. the set of all points that have an open neighbourhood entirely contained in  $X \setminus A$ .
3. The **boundary**  $\partial A$  of  $A$  is the set of all points  $x \in X$  lying neither in  $\text{int } A$ , nor in  $\text{ext } A$ .

The interior of  $A$  is occasionally written as  $\overset{\circ}{A}$ . Intuitively, the interior is the largest open subset of  $A$ , and the exterior is the largest open set not containing any point of  $A$ .

**Exercise 2.19.** Show that:

1. The interior of any set is open.
2. The exterior of any set is open.
3. The interior of  $A$  is the union of all open subsets of  $A$ .
4. The exterior of  $A$  is the union of all open subsets of  $X \setminus A$ .
5. The boundary of  $A$  is the set of all points  $x \in X$  such that every neighbourhood of  $x$  contains both points in  $A$  and points in  $X \setminus A$ .

#### Definition 2.20: Closure

The closure of  $A$  is the set  $\overline{A} := A \cup \partial A$  containing  $A$  and all of its boundary points.

**Exercise 2.21.**

1. Show that  $\overline{A}$  is the set of all points  $x \in X$  such that every neighbourhood of  $x$  contains points in  $A$ .
2. Show that  $\overline{A}$  is closed.
3. Show  $\overline{A}$  is the intersection of all closed sets containing  $A$ .

**Theorem 2.22.** Let  $B \subseteq \mathbb{R}$  be a bounded subset of  $\mathbb{R}$ . Then  $\inf B$  and  $\sup B$  are contained in  $\overline{B}$ .

*Proof.* Since  $B$  is bounded, completeness of the real numbers tells us that  $\inf B$  and  $\sup B$  are finite real numbers. If  $\inf B \notin B$ , then there would exist an open neighbourhood  $U_{\inf}$  of  $\inf B$  not containing any points of  $B$ . Since any neighbourhood of a real number  $x$  must contain numbers greater than  $x$ , this would give us a lower bound of  $B$  greater than  $\inf B$ , which contradicts the definition of  $\inf B$ . Similarly, if  $\sup B \notin \overline{B}$ , we could get a lesser upper bound of  $B$ , which is again a contradiction.  $\square$

## 2.3 Convergence

### Theorem 2.23: Detecting openness and closedness using sequences

Let  $M$  be a metric space. Let  $A \subseteq M$ . Let  $x \in M$ . Then:

1.  $x \in \text{int } A$  if and only if every sequence in  $M$  converging to  $x$  is eventually in  $A$ .
2.  $x \in \overline{A}$  if and only if there exists a sequence in  $A$  converging to  $x$ .

*Proof.* 1. If  $x \in \text{int } A$ , then there exists an  $\varepsilon > 0$  such that  $B_\varepsilon(x) \subseteq A$ . If a sequence converges to  $x$ , it must thus be eventually in  $B_\varepsilon(x)$ , and thus eventually in  $A$ .

Vice versa, if  $x \notin \text{int } A$ , then every open ball  $B_n := B_{\frac{1}{n}}(x)$  around  $A$  contains at least one point not in  $A$ , and so we can construct a sequence converging to  $x$  that isn't eventually in  $A$  by picking these points.

2. If there exists sequence of points in  $A$  converging to  $x$ , then every neighbourhood of  $x$  must contain at least one point in  $A$ , so  $x \in \overline{A}$ .

Vice versa, if  $x \in \overline{A}$ , then every open ball around  $x$  must contain a point from  $A$ , so we can construct a sequence in  $A$  converging to  $x$  by taking the sequence  $B_n := B_{\frac{1}{n}}(x)$  of open balls and picking a point  $x_n \in A \cap B_n$  for each  $n \in \mathbb{N}$ .

□

### Corollary 2.24.

1.  $A$  is closed in  $X$  if and only if it contains every limit of every convergent sequence of points in  $A$ .
2.  $A$  is open in  $X$  if and only if every sequence in  $X$  converging to a point of  $A$  is eventually in  $A$ .

## 2.4 Continuity

**Definition 2.25.** Let  $f : M \rightarrow N$  be a function between metric spaces. Then we call  $f$  *uniformly continuous* if it is continuous and fulfills the additional requirement that, given an  $\varepsilon$ , we can pick the same  $\delta$  for all points  $x \in M$ , i.e:

$$\forall \varepsilon \in \mathbb{R}_{>0} :$$

$$\exists \delta \in \mathbb{R}_{>0} :$$

$$\forall x, y \in M :$$

$$d(x, y) < \delta \implies d(f(x), f(y)) < \varepsilon$$

## 2.5 Compactness

**Exercise 2.26.** Any closed subset of a compact set is compact.

**Exercise 2.27.** A compact subset of a metric space is always closed.

### Theorem 2.28: Compactness in Metric Spaces

1. A subset of a metric space is compact if and only if it is sequentially compact.
2. A compact subset of a metric space is closed and bounded.

*Proof.* Let  $K \subseteq M$  be compact. Since  $M$  is a metric space,  $K$  is closed. If we cover  $K$  using open balls, that cover must have a finite subcover. Thus,  $K$  must be contained in a finite union of open balls, which means  $K$  must be bounded.  $\square$

### Theorem 2.29

Let  $f : M \rightarrow N$  be a continuous function between metric spaces. Let  $K \subseteq M$  be compact. Then  $f(K)$  is compact.

*Proof.* Let  $K \subseteq M$  be compact. Let  $\mathcal{U}$  be an open cover of  $f(K)$ . Since  $f$  is continuous,  $f^{-1}[\mathcal{U}] := \{f^{-1}[U] \mid U \in \mathcal{U}\}$  is an open cover of  $K$ , which must have a finite subcover  $\mathcal{V} \subseteq f^{-1}[\mathcal{U}]$ . Since  $f : K \rightarrow f(K)$  is surjective by definition,  $f[\mathcal{V}]$  must be a finite cover of  $f(K)$ .  $\square$

### Corollary 2.30: Maximum-Minimum-Theorem

Let  $f : K \rightarrow \mathbb{R}$  be a function from a compact subset of a metric space into the real numbers. Then  $f$  attains a maximum and a minimum.

*Proof.* By the previous theorem,  $f(K)$  is a compact subset of  $\mathbb{R}$ , i.e. closed and bounded by Theorem 2.28. But then Theorem 2.22 tells us that  $\inf f(K)$  and  $\sup f(K)$  are already contained in  $\overline{f(K)} = f(K)$ .  $\square$

**Lemma 2.31.** Let  $M, N$  be metric spaces. Let  $K \subseteq M$  be compact. Let  $f : K \rightarrow N$  be continuous. Then the preimage of any compact subset of  $N$  is a compact subset of  $K$ .

*Proof.* Since  $N$  is a metric space, compact subsets are closed. Since  $f$  is continuous, the preimage of a closed subset is closed. Since  $K$  is compact, any closed subset of  $K$  is compact.  $\square$

### Theorem 2.32: Countable Metric Tychonoffs Theorem

Let  $M_1, M_2$  be metric spaces. Let  $K_1 \subseteq M_1, K_2 \subseteq M_2$  be compact. Then  $K_1 \times K_2 \subseteq M_1 \times M_2$  is compact.

*Proof.* Using whichever bijection  $\mathbb{N}^2 \cong \mathbb{N}$  the reader prefers, we can unroll a sequence  $s_n$  in  $K_1 \times K_2$  into a sequence  $x_n$  in  $K_1$  and a sequence  $y_n$  in  $K_2$ , such that the elements of  $s_n$  are in one-to-one correspondence with pairs  $(x_i, y_j)$  of elements of  $x_n$  and  $y_n$ . Since  $K_1$  and  $K_2$  are compact,  $x_n$  and  $y_n$  have convergent subsequences. Using our bijection, these can be reassembled into a subsequence of  $s_n$ , and since both components of this subsequence converge, the subsequence itself converges. Thus, every sequence in  $K_1 \times K_2$  has a convergent subsequence. Since sequential compactness and compactness are equivalent in metric spaces by Theorem 2.28,  $K_1 \times K_2$  is compact.  $\square$

### Theorem 2.33

Let  $M, N$  be metric spaces. Let  $K \subseteq M$  be compact. Then every continuous function  $f : K \rightarrow N$  is uniformly continuous.

*Proof.* Assume that  $f : K \rightarrow N$  isn't uniformly continuous. Then there exists an  $\varepsilon \in \mathbb{R}_{>0}$  such that for every  $n \in \mathbb{N}$ , we can find points  $x_n, y_n \in K$  such that  $d(x_n, y_n) < 1/n$ , but  $d(f(x_n) - f(y_n)) \geq \varepsilon$ .

Since  $K$  is compact, it is sequentially compact by Theorem 2.28, so there exists a subsequence  $(x_k) \subset (x_n)$  converging to a point  $p$ . Since  $d(x_n, y_n) < 1/n$ , there also exists a subsequence  $x_k \subset y_n$  converging to  $p$ . If  $f$  was continuous, we would have

$$\lim_{k \rightarrow \infty} f(x_k) = \lim_{k \rightarrow \infty} f(y_k),$$

contradicting the fact that we have  $d(f(x_k) - f(y_k)) \geq \varepsilon$  for all  $k \in \mathbb{N}$ .  $\square$

## 2.6 Connectedness

## **2.7 Banach's Fixed Point Theorem**

## Chapter 3

# Infinite Series

3.1 Achilles and the Turtle

3.2 Convergence Tests

3.3 Euler's Number  $e$

3.4 The Exponential Function

## Chapter 4

# Euclidean Geometry

### 4.1 The Complex Numbers $\mathbb{C}$

### 4.2 Circles, Angles and Trigonometry

## Chapter 5

# Function Spaces

Since we know that a sequence in a finite cartesian product converges if and only if all components converge, it seems natural to generalize this notion to arbitrary, potentially infinite cartesian products, i.e. arbitrary functions:

### Definition 5.1: Pointwise Convergence

We say that a sequence of functions  $f_n : M \rightarrow N$  converges *pointwise* to a given function  $f : M \rightarrow N$  if and only if all components  $f_n(x)$  converge in the usual sense, i.e:

$$\begin{aligned} &\forall x \in M : \\ &\forall \varepsilon \in \mathbb{R}_{>0} : \\ &\exists N \in \mathbb{N} : \\ &\forall n \geq N : \\ &\quad d(f_n(x) - f(x)) < \varepsilon \end{aligned}$$

We can't put a metric on  $N^M$  in which this form of converges matches regular metric convergence, but the intuition that this is the most natural form of convergence in  $N^M$  will be vindicated once we study abstract product spaces in point-set topology, at which point we confirm that in the natural product topology, an infinite cartesian product still converges if and only if every component converges individually.

However, there *is* a natural metric to put on  $N^M$ , which leads to a stronger form of convergence:

**Definition 5.2.** An *extended metric space* is a metric space where  $d$  can also take on the value  $\infty$ .

### Theorem 5.3: Supremum Metric

The set  $N^M$  of functions from a metric space  $M$  to a metric space  $N$  can be metrized by the extended metric

$$d_\infty(f, g) := \sup_{x \in M} |f(x) - g(x)|.$$

Additionally, this function is a regular metric on the set  $B(M, N) \subset N^M$  of bounded functions  $M \rightarrow N$ .

Note that is simply a direct generalization of the Chebyshev metric 2.3 to infinite cartesian products!

### Theorem 5.4: Uniform Convergence

In  $(N^M, d_\infty)$ , a sequence of functions  $f_n$  converges to a given function  $f$  if and only if

$$\begin{aligned} \forall \varepsilon \in \mathbb{R}_{>0} : \\ \exists N \in \mathbb{N} : \\ \forall n \geq N : \\ \forall x \in M : \\ d(f_n(x) - f(x)) < \varepsilon \end{aligned}$$

we call this form of convergence of a sequence of functions *uniform convergence*.

**Theorem 5.5.** Let  $U \subseteq (X, d)$  be open and path-connected. Then there exists a piecewise linear path  $\gamma_n : [0, 1] \rightarrow U$  between any two  $p, q \in U$ .

*Proof Sketch.* 1. Since  $U$  is path connected, there exists a curve  $\gamma : [0, 1] \rightarrow U$  connecting  $p$  and  $q$ .

2. We can approximate  $\gamma$  via a piecewise linear curve  $\gamma_n$  by setting

$$\gamma_n\left(\frac{x}{n}\right) := \gamma\left(\frac{x}{n}\right)$$

3. Since  $\gamma$  is continuous and defined on a compact set, it is uniformly continuous by Theorem 2.5. From this, it can be shown that the  $\gamma_n$  converge to  $\gamma$  uniformly.

4. Since  $\gamma([0, 1]) \subseteq U$  is compact and  $X \setminus U$  is closed and disjoint from  $U$ , we have  $\varepsilon := d(\gamma([0, 1]), X \setminus U) > 0$ .

5. Therefore, uniform convergence guarantees there is an  $n$  such that  $d(\gamma_n([0, 1]), \gamma([0, 1])) < \varepsilon$ , which implies  $\gamma_n([0, 1]) \subseteq U$ .

”□”

## **Part III**

# **Calculus in one Variable**

## **Chapter 1**

# **Derivatives**

## **Chapter 2**

# **Integrals**

## **Part IV**

# **Multivariable Calculus**

# Chapter 1

## Differentiation in $\mathbb{R}^n$

### 1.1 Differentials and Partial Derivatives

#### Definition 1.1: Fréchet Derivative

Let  $V, W$  be normed vector spaces. Let  $U$  be an open subset of  $V$ . Then  $f : U \rightarrow W$  is called *Fréchet differentiable* or just *differentiable* at  $p \in U$  if there exists a continuous linear map  $Df(p) : V \rightarrow W$  such that

$$\lim_{\|h\|_V \rightarrow 0} \frac{\|f(p+h) - f(p) - Df(p)(h)\|_W}{\|h\|_V} = 0.$$

We call  $Df(p)$  the *differential of  $f$  at  $p$* . We call  $f$  *differentiable* if it is differentiable at every point  $p \in U$ . We call  $f$  *continuously differentiable* if the map  $Df : p \mapsto Df(p)$  is continuous.

Recall that linear maps are continuous if and only if they are bounded, which is always the case if  $V$  is finite-dimensional.

**Exercise 1.2.** Verify that if we identify scalars  $c \in \mathbb{R}$  with linear maps  $x \mapsto c \cdot x$ , this definition agrees with the usual definition for  $f : \mathbb{R} \rightarrow \mathbb{R}$ .

**Theorem 1.3.** If  $f : U \rightarrow W$  is differentiable, it is continuous.

*Proof.* If  $f$  is differentiable, we have

$$\lim_{\|h\|_V \rightarrow 0} \frac{\|f(p+h) - f(p) - Df(p)(h)\|_W}{\|h\|_V} = 0.$$

This can only be the case if

$$\lim_{\|h\|_V \rightarrow 0} \|f(p+h) - f(p) - Df(p)(h)\|_W = 0,$$

i.e. if

$$\lim_{\|h\|_V \rightarrow 0} f(p+h) = \lim_{\|h\|_V \rightarrow 0} f(p) + Df(p)(h).$$

Since  $Df(p)$  is continuous with  $Df(0) = 0$ , this simplifies to

$$\lim_{\|h\|_V \rightarrow 0} f(p+h) = f(p),$$

which is the definition of sequential continuity.  $\square$

### Definition 1.4: Directional and Partial Derivates

Let  $V$  and  $W$  be normed vector spaces. Let  $v \in V$ . Let  $U \subseteq V$  be open and  $f : U \rightarrow W$ . Then we call the limit

$$\frac{\partial}{\partial v} f(p) = \left. \frac{d}{dh} f(p + hv) \right|_{h=0} = \lim_{h \rightarrow 0} \frac{f(p + hv) - f(p)}{h}$$

the **directional derivative** of  $f$  at  $p$  in direction  $v$ . If  $(x_0, \dots, x_n)$  is a basis of  $V$ , we call the directional derivatives

$$\partial_i f(p) := \frac{\partial}{\partial x_i} f(p)$$

the **partial derivatives of  $f$  at  $p$** . If all partial derivatives exist, we call  $f$  **partially differentiable**. If all maps  $\partial_i f : p \mapsto \partial_i f(p)$  are continuous, we call  $f$  **continuously partially differentiable**.

Partial derivatives aren't actually any more complicated than standard one-dimensional derivatives - you can calculate partial derivatives by simply fixing all coordinates except one and then taking a regular derivative in the remaining coordinate. For example, if we have a function

$$\begin{aligned} f : \mathbb{R}^3 &\rightarrow \mathbb{R} \\ (x, y, z) &\mapsto f(x, y, z), \end{aligned}$$

we can define a function  $f|_1 : \mathbb{R} \rightarrow \mathbb{R}$  by restriction to the first coordinate:

$$f|_1(x) := f(x, y, z).$$

Using this function, we can simplify the partial derivative:

$$\begin{aligned} \partial_1 f(x, y, z) &= \lim_{h \rightarrow 0} \frac{f((x, y, z) + he_1) - f(x, y, z)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x + h, y, z) - f(x, y, z)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f|_1(x + h) - f|_1(x)}{h} \\ &= \frac{d}{dx} f|_1(x) \end{aligned}$$

This tells us that, for example, the function  $f(x, y, z) = x \cdot y \cdot z$  has partial derivatives

$$\begin{aligned} \partial_1 f(x, y, z) &= \frac{d}{dx} xyz = yz \\ \partial_2 f(x, y, z) &= \frac{d}{dy} xyz = xz \\ \partial_3 f(x, y, z) &= \frac{d}{dz} xyz = xy. \end{aligned}$$

**Exercise 1.5.** 1. Show that if  $f$  is differentiable, then all directional derivatives exist and fulfill  $\frac{\partial}{\partial v} f(p) = Df(p)(v)$ .

2. Conclude that  $Df(p)$  is represented in basis  $(x_0, \dots, x_n)$  by the **Jacobian matrix**

$$Jf(p) = \left( \frac{\partial}{\partial x_1} f(p), \dots, \frac{\partial}{\partial x_n} f(p) \right) = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(p) & \dots & \frac{\partial}{\partial x_n} f_1(p) \\ \vdots & & \vdots \\ \frac{\partial}{\partial x_1} f_n(p) & \dots & \frac{\partial}{\partial x_n} f_n(p) \end{pmatrix}$$

If the image of  $f$  is one-dimensional, the Jacobian matrix is simply a row vector. We call the transpose of this vector the **gradient of  $f$  at  $p$**  and denote it

$$\nabla f(p) := Jf(p)^\top = \begin{pmatrix} \frac{\partial}{\partial x_1} f(p) \\ \vdots \\ \frac{\partial}{\partial x_n} f(p) \end{pmatrix}$$

The transposition comes from the fact that we would like to be able to get the directional derivatives of  $f$  at  $p$  by taking a scalar product with the gradient, which gives us

$$\begin{aligned} \langle \nabla f(p), v \rangle &= \nabla f(p)^\top v \\ &= Jf(p) \cdot v \\ &= Df(p)(v) \\ &= \frac{\partial}{\partial v} f(p) \end{aligned}$$

There is one very important pitfall to watch out for when it comes to differentiability: We have just shown that if  $f$  is differentiable, we can get the differential by construction the Jacobian matrix from the partial derivatives. This makes it look like every partially differentiable function is differentiable, which turns out to not be true in general.

For example, there exist functions which are partially differentiable, but not continuous - we showed earlier that all differentiable functions are continuous, so these functions are quick counterexamples.

One example is

$$\begin{aligned} f : \mathbb{R}^2 &\rightarrow \mathbb{R} \\ (x, y) &\mapsto \begin{cases} \frac{xy}{(x^2+y^2)^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases} \end{aligned}$$

This function has  $\partial_1 f(0) = \partial_2 f(0) = 0$ , since  $f(h, 0) = f(0, h) = 0$  for all  $h \in \mathbb{R}$ , but isn't continuous, since  $\lim_{n \rightarrow \infty} f\left(\frac{1}{n}, \frac{1}{n}\right) = \lim_{n \rightarrow \infty} \frac{n^2}{4} = \infty \neq f(0)$ .

For a more involved counterexample, consider the function

$$\begin{aligned} f : \mathbb{R}^2 &\rightarrow \mathbb{R} \\ (x, y) &\mapsto \begin{cases} \frac{y^3}{x^2+y^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases} \end{aligned}$$

This function is continuous - for  $(x, y) \neq 0$ , it is a composition of continuous

functions, and thus continuous, and since  $x^2 + y^2 \geq y^2$ , we additionally have

$$\begin{aligned} |f(x, y)| &= \frac{|y^3|}{|x^2 + y^2|} \\ &= \frac{|y^3|}{x^2 + y^2} \\ &\leq \frac{|y^3|}{y^2} \\ &= |y|, \end{aligned}$$

so  $f(x, y)$  converges to  $f(0, 0) = 0$  as  $(x, y)$  converges to 0, implying that  $f$  is also continuous at the origin.

$f$  is also directionally differentiable in all directions:

$$\begin{aligned} \partial_v f(0) &= \lim_{h \rightarrow 0} \frac{f(h \cdot v) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h \cdot v)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\left( \frac{h^3 v_2^3}{h^2 v_1^2 + h^2 v_2^2} \right)}{h} \\ &= \lim_{h \rightarrow 0} \frac{h^3 v_2^3}{h^3 v_1^2 + h^3 v_2^2} \\ &= \lim_{h \rightarrow 0} \frac{v_2^3}{v_1^2 + v_2^2} \\ &= \frac{v_2^3}{v_1^2 + v_2^2} \\ &= f(v) \end{aligned}$$

In particular,  $\partial_1 f(0) = 0$  and  $\partial_2 f(0) = 1$ . However,  $f$  is not differentiable at 0, since otherwise our representation  $Df(0) = Jf(0)$  would give us:

$$\begin{aligned} \lim_{\vec{h} \rightarrow 0} \frac{|f(h) - f(0) - Df(0)(\vec{h})|}{\|\vec{h}\|_2} &= \lim_{\vec{h} \rightarrow 0} \frac{|f(h) - Df(0)(h)|}{\|\vec{h}\|_2} \\ &= \lim_{\vec{h} \rightarrow 0} \left| \frac{f(h) - \partial_1 f(0) \cdot h_1 + \partial_2 f(0) \cdot h_2}{\sqrt{h_1^2 + h_2^2}} \right| = \lim_{\vec{h} \rightarrow 0} \left| \frac{f(h) - h_2}{\sqrt{h_1^2 + h_2^2}} \right| \\ &= \lim_{\vec{h} \rightarrow 0} \left| \frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left( \frac{h_2^3}{h_1^2 + h_2^2} - h_2 \right) \right| = \lim_{\vec{h} \rightarrow 0} \left| \frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left( \frac{h_2^3 - h_2(h_1^2 + h_2^2)}{h_1^2 + h_2^2} \right) \right| \\ &= \lim_{\vec{h} \rightarrow 0} \left| \frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left( \frac{h_2^3 - h_2 h_1^2 - h_2^3}{h_1^2 + h_2^2} \right) \right| = \lim_{\vec{h} \rightarrow 0} \left| \frac{1}{\sqrt{h_1^2 + h_2^2}} \cdot \left( \frac{-h_2 h_1^2}{h_1^2 + h_2^2} \right) \right| = \lim_{\vec{h} \rightarrow 0} \left| \frac{h_2 h_1^2}{\sqrt{h_1^2 + h_2^2}^3} \right| \end{aligned}$$

Since we are only interested in whether this expression converges to 0, we can

square it to simplify slightly:

$$\left| \frac{h_2 h_1^2}{\sqrt{h_1^2 + h_2^2}^3} \right|^2 = \left| \frac{h_2^2 h_1^4}{(h_1^2 + h_2^2)^3} \right|$$

If we now set  $h := (\frac{1}{n}, \frac{1}{n})$ , we get:

$$\left| \frac{h_2^2 h_1^4}{(h_1^2 + h_2^2)^3} \right| = \left| \frac{\frac{1}{n^6}}{\left(\frac{2}{n^2}\right)^3} \right| = \left| \frac{\frac{1}{n^6}}{\frac{8}{n^6}} \right| = \frac{1}{8} \neq 0$$

The important takeaway here is that directional differentiability only guarantees "differentiability along straight paths", while full differentiability also requires "differentiability along curved paths".

### Theorem 1.6

$f : V \rightarrow W$  is continuously differentiable if and only if it is continuously partially differentiable.

*Proof.*  $\Rightarrow$ : We have  $\text{Hom}(\mathbb{R}^n, \mathbb{R}^m) \cong \mathbb{R}^{mn}$ . Therefore, the map  $Jf : \mathbb{R}^n \mapsto \mathbb{R}^{mn}$  is continuous if and only if every component  $(Jf)_i : \mathbb{R}^n \mapsto \mathbb{R}$  is continuous. By the definition of  $Jf$ , these are the partial derivatives.

$\Leftarrow$ : Since every normed complex vector space is isomorphic to a normed real vector space, we can assume without loss of generality that  $W$  is real. Since all norms are equivalent, and since a sequence in  $W$  converges iff. all components converge, we can additionally assume that  $W$  is one-dimensional, i.e.  $W \cong \mathbb{R}$ .

We need to show that

$$\lim_{h \rightarrow \vec{0}} \frac{f(p+h) - f(p) - Jf(p)h}{\|h\|_2} = \vec{0}$$

Let  $h := \sum_{i=1}^n h_i e_i$ . We decompose the path from  $p$  to  $p+h$  into its partial steps, taking one coordinate direction at a time:

$$\forall k \in [0..n] :$$

$$p_k = p + \sum_{i=1}^k h_i e_i$$

Now, by the mean value theorem, there exists an  $s_k \in [0, 1]$  such that

$$\begin{aligned} f(p_k) - f(p_{k-1}) &= f(p_{k-1} + h_k e_k) - f(p_{k-1}) \\ &= \partial_k f(p_{k-1} + s_k h_n e_n) h_k \end{aligned}$$

Now, since

$$Jf(p)(h) = \sum_{i=1}^n \partial_i f(p) h_i,$$

we get:

$$\begin{aligned}
& \frac{|f(p+h) - f(p) - Jf(p)(h)|}{\|h\|_2} \\
&= \frac{1}{\|h\|_2} \left| f(p_n) - f(p_0) - \sum_{i=1}^n \partial_i f(p) h_i \right| \\
&= \frac{1}{\|h\|_2} \left| \sum_{i=1}^n f(p_i) - f(p_{i-1}) - \partial_i f(p) h_i \right| \\
&= \frac{1}{\|h\|_2} \left| \sum_{i=1}^n \partial_i f(p_{i-1} + s_i h_i e_i) h_i - \partial_i f(p) h_i \right| \\
&= \frac{1}{\|h\|_2} \left| \sum_{i=1}^n (\partial_i f(p_{i-1} + s_i h_i e_i) - \partial_i f(p)) h_i \right| \\
&= \frac{1}{\|h\|_2} \left| \sum_{i=1}^n \left( \partial_i f \left( p + \sum_{j=1}^{i-1} h_j e_j + s_i h_i e_i \right) - \partial_i f(p) \right) h_i \right| \\
&\leq \frac{1}{\|h\|_2} \sum_{i=1}^n \left| \left( \partial_i f \left( p + \sum_{j=1}^{i-1} h_j e_j + s_i h_i e_i \right) - \partial_i f(p) \right) h_i \right| \\
&= \sum_{i=1}^n \left| \partial_i f \left( p + \sum_{j=1}^{i-1} h_j e_j + s_i h_i e_i \right) - \partial_i f(p) \right| \cdot \frac{|h_i|}{\|h\|_2} \\
&\leq \sum_{i=1}^n \left| \partial_i f \left( p + \sum_{j=1}^{i-1} h_j e_j + s_i h_i e_i \right) - \partial_i f(p) \right|
\end{aligned}$$

Since  $s_i \in [0, 1]$ , we have

$$\lim_{h \rightarrow 0} p + \sum_{j=1}^{i-1} h_j e_j + s_i h_i e_i = p$$

And thus, since the partial derivatives  $\partial_i f$  are continuous, the right side converges to zero, so  $f$  is differentiable.  $\square$

**Exercise 1.7.** Let  $V$  be an inner product space over a field  $\mathbb{F} = \mathbb{R}$  or  $\mathbb{F} = \mathbb{C}$ . Let  $v \in V$ . Then the function

$$\begin{aligned}
\langle \cdot, v \rangle : V &\rightarrow \mathbb{F} \\
w &\mapsto \langle w, v \rangle
\end{aligned}$$

is differentiable, with  $D \langle \cdot, v \rangle (w) = v^\top$  for all  $w \in V$ .

**Exercise 1.8.** Let  $b$  be a symmetric bilinear form. Let

$$q(v) := b(v, v)$$

be the corresponding quadratic form. Then  $q$  is differentiable, with

$$Dq(v)(w) = 2b(v, w)$$

## 1.2 Rules of Differentiation

**Exercise 1.9.** Show that the differential operator  $D$  is linear, i.e. that we have

$$D(\alpha f + \beta g)(p) = \alpha Df(p) + \beta Dg(p)$$

for  $f, g$  differentiable at  $p$ .

### Theorem 1.10: Chain Rule in $\mathbb{R}^n$

Let  $f : U \rightarrow \mathbb{R}^m$  and  $g : V \rightarrow \mathbb{R}^r$  with  $U \subset \mathbb{R}^n, V \subset \mathbb{R}^m$  open and  $f(U) \subseteq V$ . Let  $f$  be differentiable at a point  $p \in U$  and let  $g$  be differentiable at  $f(p)$ . Then

$$D(g \circ f)(p) = Dg(f(p)) \circ Df(p)$$

**Exercise 1.11.** Verify that this version of the chain rule agrees with the familiar chain rule for  $f, g : \mathbb{R} \rightarrow \mathbb{R}$ .

*Proof.* By definition of the differential, we have:

$$\begin{aligned} f(x) &= f(p) \\ &\quad + Df(p)(x - p) \\ &\quad + \varepsilon_f(x)\|x - p\| \\ g(f(x)) &= g(f(p)) \\ &\quad + Dg(f(p))(f(x) - f(p)) \\ &\quad + \varepsilon_g(f(x))\|f(x) - f(p)\| \end{aligned}$$

For some functions  $\varepsilon_f : U \rightarrow \mathbb{R}^m, \varepsilon_g : U \rightarrow \mathbb{R}^r$  which are continuous in  $p$  and  $f(p)$  and fulfill  $\varepsilon_f(p) = 0$  and  $\varepsilon_g(f(p)) = 0$ .

The second equation tells us:

$$\begin{aligned} g(f(x)) &= g(f(p)) \\ &\quad + Dg(f(p))(f(x) - f(p)) \\ &\quad + \varepsilon_g(f(x))\|f(x) - f(p)\| \end{aligned}$$

Plugging in our equation for  $f(x)$  now lets us regroup our terms into an

equation of the same form as the definition of the differential:

$$\begin{aligned}
g(f(x)) &= g(f(p)) \\
&\quad + Dg(f(p)) \left[ \overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\| - f(p)}^{f(x)} \right] \\
&\quad + \varepsilon_g(f(x)) \left\| \overbrace{f(p) + Df(p)(x-p) + \varepsilon_f(x)\|x-p\| - f(p)}^{f(x)} \right\| \\
&= g(f(p)) \\
&\quad + Dg(f(p)) \left[ Df(p)(x-p) + \varepsilon_f(x)\|x-p\| \right] \\
&\quad + \varepsilon_g(f(x)) \left\| Df(p)(x-p) + \varepsilon_f(x)\|x-p\| \right\| \\
&= g(f(p)) \\
&\quad + Dg(f(p))Df(p)(x-p) \\
&\quad + Dg(f(p))\varepsilon_f(x)\|x-p\| \\
&\quad + \varepsilon_g(f(x)) \left\| Df(p)(x-p) + \varepsilon_f(x)\|x-p\| \right\| \\
&= g(f(p)) \\
&\quad + Dg(f(p))Df(p)(x-p) \\
&\quad + \underbrace{\left( \varepsilon_f(x)Dg(f(p)) + \varepsilon_g(f(x)) \left\| Df(p) \left( \frac{x-p}{\|x-p\|} \right) + \varepsilon_f(x) \right\| \right)}_{:=\varepsilon_{g \circ f}(x)} \|x-p\| \\
&= g(f(p)) \\
&\quad + (Dg(f(p)) \circ Df(p))(x-p) \\
&\quad + \varepsilon_{g \circ f}(x)\|x-p\|
\end{aligned}$$

All that remains is to show that  $\varepsilon_{g \circ f}(x)$  is continuous if we set

$$\varepsilon_{g \circ f}(p) = 0.$$

Since we have  $\varepsilon_f(p) = 0$  and  $\varepsilon_g(f(p)) = 0$ , we have our desired result if we can show that

$$\left\| Df(p) \left( \frac{x-p}{\|x-p\|} \right) \right\| = \frac{\|Df(p)(x-p)\|}{\|x-p\|}.$$

is bounded. Since  $\mathbb{R}^n$  is finite-dimensional, we can find a constant  $c$  such that

$$\|Df(p)(x-p)\| \leq c \cdot \|x-p\|,$$

which means that our quotient is indeed bounded, completing the proof.  $\square$

**Theorem 1.12: Product Rule in  $\mathbb{R}^n$** 

Let  $U \subseteq \mathbb{R}^n$ . Let  $f, g : U \rightarrow \mathbb{R}$  at a point  $p \in U$ . Then

$$D(f \cdot g)(p) = Df(p) \cdot g(p) + f(p) \cdot Dg(p).$$

**Exercise 1.13.** Verify that this version of the product rule agrees with the familiar product rule for  $f, g : \mathbb{R} \rightarrow \mathbb{R}$ .

*Proof.* We define the multiplication function  $m(x, y) := x \cdot y$ . The partial derivatives are  $\partial_1 m(x, y) = y$  and  $\partial_2 m(x, y) = x$ , i.e.

$$Dm(x, y) = \begin{pmatrix} y & x \end{pmatrix}$$

Applying the chain rule now gives

$$\begin{aligned} D(f \cdot g)(p) &= D(m \circ (f, g))(p) \\ &= Dm((f, g)(p)) \circ D(f, g)(p) \\ &= Dm(f(p), g(p)) \circ (Df(p), Dg(p)) \\ &= Dm(f(p), g(p)) \cdot (Df(p), Dg(p)) \\ &= \begin{pmatrix} g(p) & f(p) \end{pmatrix} \cdot \begin{pmatrix} Df(p) \\ Dg(p) \end{pmatrix} \\ &= Df(p) \cdot g(p) + f(p) \cdot Dg(p). \end{aligned}$$

□

**Exercise 1.14.** Let  $f : V \rightarrow W$  be differentiable. Let  $\gamma : [0, 1] \rightarrow V$  be differentiable. Then  $f \circ \gamma$  is differentiable with

$$(f \circ \gamma)'(t) = Df(\gamma'(t)) \cdot \gamma'(t)$$

**Theorem 1.15.** Let  $U \subseteq V$  be open. Let  $f : U \rightarrow W$  be differentiable with  $Df = 0$ . Then  $f$  is locally constant.

**Corollary 1.16.** Let  $U \subseteq V$  be open and connected. Let  $f : U \rightarrow W$  be differentiable with  $Df = 0$ . Then  $f$  is constant.

## 1.3 Higher Partial Derivatives

### Theorem 1.17: Schwartz's Theorem

Let  $f : U \subseteq V \rightarrow W$  be twice continuously partially differentiable. Then the order of partial derivation does not matter, i.e.

$$\partial_i \partial_j f = \partial_j \partial_i f$$

for all  $i, j \in [1..n]$ .

### Theorem 1.18: First-Order Taylor Expansion in multiple Variables

Let  $f : U \subseteq V \rightarrow W$ . Let  $f$  be continuously differentiable. Let  $x \in U$ ,  $h \in V$  fulfill  $[x, x + h] \subseteq U$ . Then

$$f(x + h) = f(x) + Df(x)(h) + r(h),$$

with  $r \in o(\|h\|)$ .

## 1.4 Differential Operators

## Chapter 2

# The Two Queens of Multivariable Calculus

### 2.1 The Inverse Function Theorem

**Theorem 2.1: Inverse Function Theorem**

Let  $U \subseteq \mathbb{R}^n$  be open. Let  $f : U \rightarrow \mathbb{R}^n$  be continuously differentiable. Let  $x \in U$  be a point such that  $Df(x) : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is invertible. Then there exists an open neighbourhood  $V$  of  $x$  such that:

1.  $f(V)$  is an open neighbourhood of  $f(x)$ .
2.  $f|_V$  is a  $C^1$ -diffeomorphism, i.e. invertible with continuously differentiable inverse.

In short: if a continuously differentiable function has an invertible differential at a given point, there exists an open neighbourhood around that point in which the function is invertible:

**Corollary 2.2.** Let  $U \subseteq \mathbb{R}^n$  be open. Let  $f : U \rightarrow \mathbb{R}^n$  be continuously differentiable. Let  $Df(x)$  be invertible for all  $x \in \mathbb{R}^n$ . Then  $f(U)$  is open.

## 2.2 The Implicit Function Theorem

### Theorem 2.3: Implicit Function Theorem

Let  $U \subseteq \mathbb{R}^m \times \mathbb{R}^k$  be open. Let  $f : U \rightarrow \mathbb{R}^k$  be continuously differentiable. Let  $(x, y) \in U$  with  $f(x, y) = z$ . Let  $Df|_{\mathbb{R}^k}(x, y) : \mathbb{R}^k \rightarrow \mathbb{R}^k$  be invertible. Then there exist

1. open neighbourhoods  $V$  of  $x$  and  $W$  of  $y$ ,
2. and a continuously differentiable function  $g : V \rightarrow W$ ,

such that the set of values in  $V \times W$  for which  $f$  takes the value  $z$  is the graph of  $g$ , i.e.

$$\{(v, w) \in V \times W \mid f(v, w) = z\} = \{(v, g(v)) \mid v \in V\}$$

In short: under mild assumptions, the level sets of a continuously differentiable function are graphs of continuously differentiable functions:

## Chapter 3

# Multiple Integrals

**Theorem 3.1.** Let  $K \subseteq \mathbb{R}^n$  be compact. Let  $f : K \times [a, b] \rightarrow \mathbb{R}$  be continuous. Then the function

$$x \mapsto \int_a^b f(x, t) dt$$

is continuous.

*Proof.*  $K \times [a, b]$  is a finite product of compact sets, i.e. compact. □

## **Chapter 4**

# **Curves**

## **Chapter 5**

# **Surfaces**